

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB	Benchmark	FM	Eval	Evidence
1	UC Merced	2010	C	21	RGB	2.1K	256	0.3m	0.317	Yes	?	TG	
2	VEDAI	2015	C	9	IRGB	1.2K	1024	0.125m	3.99	Yes	?	Paper	
3	NWPU-RESISC45	2016	C	45	RGB	31.5K	256	0.2~30m	0.405	Yes	?	Paper+TG	
4	AID	2017	C	30	RGB	10K	600	3m	2.44	Yes	Yes	Paper+REO	
5	RSI-CB128	2017	C	45	RGB	36K	128	0.3~3m	0.879	Yes	?	Paper	
6	RSI-CB256	2017	C	35	RGB	24K	256	0.3~3m	2.24	Yes	?	Paper	
7	Cactus Aerial Photos	2018	C	2	RGB	24K	32	—	0.053	Yes	Yes	FOMO	
8	DeepGlobe-Bldng	2018	S,C	1	OPT (Satellite)	24.6K	650	—	—	Yes	?	Paper	
9	EuroSAT	2018	C	10	S2 MS	27K	64	10~60m	1.92	Yes	Yes	GFM+Paper+TG	
10	FMoW	2018	C	63	RGB+MS	523.8K	—	0.3m	3.5K	Yes	Yes	Paper+GFM	
11	PatternNet	2018	C	38	RGB	30.4K	256	0.062~4.693m	1.3	Yes	?	Paper+TG	
12	xView	2018	C	60	RGB+MS	1.4K	3000	0.3m	20	Yes	?	Paper	
13	BigEarthNet-MM	2019	C	19	S1/S2	590.3K	20~120	10~60m	121	Yes	Yes	FOMO+GFM+Paper	
14	SEN12MS	2019	S,C	—	MS+SAR	180.7K	256	10m	510	Yes	Yes	FOMO+TG	
15	So2Sat LCZ42	2019	C	—	S1+S2	400.7K	32	10m	51.8	Yes	Yes	CB+GB2+GFM+Paper	
16	BrazilDAM	2020	C	2	RGB	769	384	10~60m	57	Yes	?	Paper	
17	CLRS	2020	C	25	RGB	15K	256	0.26~8.85m	1.74	Yes	?	Paper	
18	IDTReeS	2020	OD,C	33	Multiple (Multiple)	591×369	20	—	0.28	Yes	?	TG	
19	MAFAT Challenge	2020	C	—	SAR	—	—	—	—	Yes	?	Name	
20	OVERHEAD MNIST	2020	C	9	Grayscale	1K	28	—	0.018	Yes	?	Paper	
21	CropHarvest	2021	C	—	Multi	95.2K	—	10~60m	22.7	Yes	?	TG	
22	Million AID	2021	C	28	RGB	1M	150~550	0.5~153m	134	Yes	?	Paper	
23	MultiScene	2021	C	36	RGB	100K	512	0.3~0.6m	6	Yes	?	Paper	
24	South Africa Crop Type	2021	C	9	S1+S2	122.7K	—	—	82.8	Yes	?	TG	
25	ZueriCrop	2021	C	48	S2	28K×71	24	10m	38.5	Yes	?	TG	
26	ADVANCE	2022	C	13	OPT (Satellite)	5.1K	256	—	4.4	Yes	?	TG	
27	AgriFieldNet	2022	C	13	MS (Satellite)	1.2K×12	256	—	0.32	Yes	?	TG	
28	CaiRoad	2022	C	—	Satellite MT	—	—	—	10.7	Yes	Yes	GFM+Paper	
29	CalFire	2022	C	—	Satellite MT	—	—	—	7.2	Yes	Yes	GFM+Paper	
30	Canadian Cropland	2022	C	10	S2	78.5K	64	10m	26	Yes	Yes	GFM	
31	fMoW-Sentinel	2022	C	62	S2 MS	882.8K	96	10~60m	—	Yes	Yes	GFM	
32	ReforesTree	2022	OD,C	6	OPT (Drone)	100	4000	—	7.5	Yes	Yes	FOMO+TG	
33	The Auto Arborist Dataset	2022	C	344	RGB+MS	2.6M	1~24	—	24	Yes	?	Paper	
34	TreeSatAI	2022	C	47	Multi	50.4K	304	0~1	16.3	Yes	Yes	FOMO+GB2+GFM+Pa	
35	Digital Typhoon	2023	C,R	8	IR	189.4K	512	5000m	—	Yes	?	TG	
36	FireRisk	2023	C	7	OPT (Satellite)	91.9K	—	—	14	Yes	?	TG	
37	MAR20	2023	C	20	RGB	3.8K	800×800	1m	1.1	Yes	?	Paper	
38	m-BigEarthNet (GEO-Bench)	2023	C	43	S2 MS	22K	120	10m	—	Yes	Yes	GB1	
39	m-Brick-Kiln (GEO-Bench)	2023	C	2	S2 MS	17,061	64	10m	1.6	Yes	Yes	GB1	
40	m-EuroSAT (GEO-Bench)	2023	C	10	S2 MS	4K	64	10m	1.2	Yes	Yes	GB1	
41	m-ForestNet (GEO-Bench)	2023	C	12	Landsat-8 MS	8,446	332	15m	1.2	Yes	Yes	GB1	
42	m-PV4GER (GEO-Bench)	2023	C	2	RGB	13,812	320	0.1m	0.68	Yes	Yes	GB1	
43	m-So2Sat (GEO-Bench)	2023	C	17	S1/S2	21,964	32	10m	—	Yes	Yes	GB1	
44	BigEarthNet v2.0 (S1+S2)	2024	C,S	19	S1/S2	549.5K pairs	120	10~60m	110	Yes	Yes	GB2+Paper	
45	PASTIS-HD	2024	S,C,I	20	Multi	—	—	1m	—	Yes	Yes	GFM	
46	PhilEO Bench	2024	R,S,C	—	S2 MS	—	—	10m	400	Yes	Yes	PH	
47	QuakeSet	2024	C,R	2	S1 SAR	3.3K	512	10m	11	Yes	?	TG	
48	SICKLE	2024	S,C	21	S1/S2+L8 TS	34.8K	—	3m-30m	—	Yes	Yes	GFM	
49	TalloS	2024	C	1160	Multi	335K	—	10m	—	Yes	Yes	FOMO	
50	BigEarthNet-L (Landsat-Bench)	2025	C	19	Landsat MS	—	—	30m	—	Yes	Yes	LB	
51	BigEarthNetV2-S2 (Copernicus-Bench)	2025	C	19	S2 MS	24K	120	10m	—	Yes	Yes	CB	
52	EuroSAT-L (Landsat-Bench)	2025	C	10	Landsat MS	—	—	30m	—	Yes	Yes	LB	
53	GEO-Bench-2	2025	C,S,R,I	—	Multi	19 datasets	—	—	—	Yes	Yes	GB2	
54	Landsat-Bench	2025	C	—	Landsat MS	3 datasets	—	30m	—	Yes	Yes	LB	
55	LC100-L (Landsat-Bench)	2025	C	—	Landsat MS	—	—	30m	—	Yes	Yes	LB	
56	REOBench	2025	C,S,OD,IC	—	OPT+Text	6 subsets	—	—	—	Yes	Yes	REO	
57	Tropical Cyclone	2020	R	—	GOES 8–16 MSI	108.1K	256x256	4K–8K	—	Yes	?	TG	
58	SustainBench Crop Yield	2021	R	—	MODIS MSI	11K	32	—	—	Yes	?	TG	
59	USAVars	2021	R	—	NAIP RGB+NIR	100K	—	4m	—	Yes	?	TG	
60	Hyperview Challenge	2022	R	—	HS	2.9K	11~250	2m	1.85	Yes	?	Name+Paper	
61	So2Sat Population	2022	R	—	DEM+S2	—	—	10m	—	Yes	?	Paper	
62	BioMassters	2023	R	—	S1/S2	13K	256	10m	201	Yes	Yes	GB2+PG+TG	
63	SKIPP'D	2023	R	—	RGB	363.4K	64	—	—	Yes	?	TG	
64	Open-Canopy	2024	R,CD	—	SPOT-6/7+LiDAR	>87K km ²	1K tiles	1.5m	—	Yes	Yes	PG+Paper	
65	Western USA Live Fuel Moisture	2024	R	—	L8+S1	2.6K	—	—	—	Yes	?	TG	
66	Air Quality	—	R	—	Multi-sensor	9.4K	—	—	—	Yes	?	TG	
67	ISPRS 2D - Potsdam	2011	S	6	RGB+nDSM	38	6000	0.05m	16	Yes	?	Paper+TG	
68	ISPRS 2D - Vaihingen	2011	S	6	RGB+nDSM	33	2200	0.09m	17	Yes	Yes	GFM+Paper+TG	
69	DFC2015 Zeebruges	2015	S	8	Multi	7	0~100	0.05m	0.002	Yes	?	Paper	
70	L8 Biome	2016	S	5	L8 MSI	96	8.9K	15~30m	96	Yes	?	TG	
71	SpaceNet-1	2016	S	1	RGB+MS	9.7K	650	0.5~1m	31	Yes	?	Paper	
72	INRIA	2017	S	1	RGB	360	1500	0.3m	19.5	Yes	?	Paper	
73	Inria Aerial Image Labeling	2017	S	2	RGB	360	5000	0.3m	—	Yes	?	Paper+TG	
74	SpaceNet-2	2017	S	1	RGB+MS	24.6K	650	0.3m	182	Yes	Yes	GB2+GFM+Paper	
75	SpaceNet-3	2017	S	1	Multi	3.7K	1300	0.3~1.24m	182	Yes	?	Paper	
76	Urban 3D Challenge	2017	S	2	RGB	157K	—	0.5m	—	Yes	?	Name	
77	DeepGlobe-LandCover	2018	S	7	RGB	1.1K	2448	0.5m	2.75	Yes	?	TG	
78	DeepGlobe-Road	2018	S	1	OPT (Satellite)	8.6K	1024	—	3.84	Yes	?	Paper	
79	DFC18	2018	S	20	Multi	1	—	0.05~1m	19.8	Yes	?	Paper	
80	DLRSD	2018	S	21	RGB	2.1K	256	0.3m	0.004	Yes	?	TG	
81	GID-15 (Gaofen Image Dataset)	2018	S	15	GF-2 RGB	150	6800×7200	3m	36	Yes	?	Paper+TG	
82	Chesapeake Land Cover	2019	S	16	RGB+MS	250K	240	1m	140	Yes	?	TG	
83	Open AI Challenge: Caribbean	2019	S	5	RGB	7	52~318	0.04m	—	Yes	?	Name+Paper	

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Table 1 continued

Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB	Benchmark	FM	Eval	Evidence
84	SpaceNet-5	2019	S	1	Multi	2.4K	1300	0.3m	84.1	Yes	?		Paper
85	BDCI2020	2020	S	7	RGB	146K	256	2m	1.3	Yes	?		Paper
86	DFC20	2020	S	17	S1+S2	180.7K	—	10m	19.2	Yes	?		Paper
87	DFC2020	2020	S	10	S1/S2	5.1K	256	10m	—	Yes	Yes		CB+GFM
88	Kenya Crop Type	2020	S	7	S2 MSI	4.7K	3035x2016	10m	—	Yes	?		TG
89	LandCoverAI	2020	S	—	RGB	41	9000	0~0.2	1.4	Yes	?		TG
90	Open Cities AI Challenge	2020	S	2	RGB	11K	1~24	0.03~0.2m	81.5	Yes	?		Name+Paper
91	SemCity Toulouse	2020	S	—	MS	16	3500	0.5~2m	5.2	Yes	?		Paper
92	Sen1Floods11	2020	S	2	S1/S2	4.8K	512	10m	14.3	Yes	Yes		GFM+PG
93	South Sudan Crop Type	2020	S	5	S1/S2+Planet	837 tiles	64	10m	977	Yes	Yes		PG
94	SpaceNet-6	2020	S	1	SAR+RGB	3.4K	900	0.5m	0.368	Yes	Yes		GB2
95	SpaceNet-7	2020	CD,S	1	MS+MT	1.5K	1024	4m	20.6	Yes	Yes		GB2+PG
96	TimeSen2Crop	2020	S	16	S2	1M	10980	10m	1.1	Yes	Yes		GFM
97	WHU-Hi	2020	S	11	HS (Drone)	3	1217	0.109m	0.817	Yes	?		Paper
98	WHU-Hi-HongHu	2020	S	22	HS	1	940x475	0.043m	—	Yes	?		Paper
99	Benin Cashews	2021	S	6	S2 (labels corrected with Pléiades)	70	1186x1122	0~1	—	Yes	?		TG
100	DFC21-DSE	2021	S	4	Multi	6K	16	10~750m	18	Yes	?		Paper
101	ETCI2021	2021	S	2	SAR (Satellite)	66.8K	256	—	5.3	Yes	?		TG
102	LoveDA	2021	S	7	RGB	6K	1024	0.3m	9.6	Yes	?		TG
103	MARIDA	2021	S	15	MS (Satellite)	1.4K×10	256	—	1.2	Yes	Yes		GFM
104	Mini Inria Aerial Image Labeling Dataset	2021	S	2	RGB	32.5K	512	0.3m	—	Yes	?		Paper
105	PASTIS	2021	S	18	S2	2433*45*10	128	10m	29	Yes	Yes		GB2+TG
106	PASTIS-R	2021	S	18	Multiple (Satellite)	2.4K	128	—	54	Yes	Yes		PG
107	Rwanda Field Boundary	2021	S	2	RGB+NIR	70	256	4.7m	—	Yes	?		TG
108	Sen4AgriNet	2021	S	—	S2	—	—	10~60m	262	Yes	?		Paper
109	CloudSEN12	2022	S	4	S2/S1+DEM	49.4K	509	10m	~1K	Yes	Yes		CB+GB2+Paper
110	DFC2022	2022	S	15	RGB	4K	2000	0.5m	—	Yes	?		TG
111	DynamicEarthNet	2022	CD,S	7	PlanetFusion	54.8K	1024	3m	524	Yes	Yes		GB2+GFM+PG
112	Five-Billion-Pixels	2022	S	24	RGB+MS	150	7200×6800	4m	104	Yes	Yes		PG
113	GeoNRW	2022	S	10	RGB+nDSM	7.8K×3	1000	1m	30	Yes	?		TG
114	MapInWild	2022	S	11	Multiple (Satellite)	1K×8	1920	—	365	Yes	?		TG
115	MDAS	2022	S	20	Multi	3	100~18K	0.3~30m	—	Yes	?		TG
116	MMFlood	2022	S	1	SAR (Satellite)	1.7K×2	2000	—	11.3	Yes	?		TG
117	MultiSenGE	2022	S	14	S1+S2	1.1M×12	256	10m	530	Yes	?		Paper
118	SeasoNet	2022	S	33	S2	1.8M	120	10m	229	Yes	?		TG
119	Sentinel-2 Cloud Cover	2022	S	2	MS	22.7K	—	10m-60m	51.2	Yes	?		TG
120	SpaceNet-8	2022	S	2	PAN+RGB	1.2K×2	1200	0.3~0.8m	6.8	Yes	Yes		GB2
121	WHU-OHS	2022	S	24	HS	7.8K	512	10m	94.9	Yes	?		Paper
122	AI4SmallFarms	2023	S	2	S2 MS	62 tiles	—	10m	—	Yes	Yes		PG
123	BraDD-S1TS	2023	CD,S	2	S1 TS	13.2K	—	10m	17.7	Yes	Yes		GFM
124	DW-Expert	2023	S	9	S1/S2	71.4K	96	10m	—	Yes	Yes		GFM
125	FLAIR #2	2023	S	13	Multi	77.8K	512	0.2m-10m	—	Yes	Yes		FOMO+GB2+GFM
126	HLS Burn Scars	2023	S	2	HLS	804	512	30m	—	Yes	Yes		GB2+GFM+PG
127	I/O Bench	2023	S	—	Landsat MSI	1	8000	30m	—	Yes	?		TG
128	m-Cashew-Plantation (GEO-Bench)	2023	S	7	S2 MS	1,800	256	10m	1.3	Yes	Yes		GB1
129	m-Chesapeake-LandCover (GEO-Bench)	2023	S	7	RGBN	5K	256	1m	—	Yes	Yes		GB1
130	m-NeonTree (GEO-Bench)	2023	S	2	RGB+HS+DEM	457	400	0.1m	—	Yes	Yes		GB1
131	m-NZ-Cattle (GEO-Bench)	2023	S	2	RGB	655	500	0.1m	0.33	Yes	Yes		GB1
132	m-PV4GER-Seg (GEO-Bench)	2023	S	2	RGB	3,806	320	0.1m	0.32	Yes	Yes		GB1
133	m-SA-Crop-Type (GEO-Bench)	2023	S	10	S2 MS	5K	256	10m	1.4	Yes	Yes		GB1
134	SSL4EO-L Benchmark	2023	S	17/134	Landsat MSI	25K	264	30m	—	Yes	?		Name+TG
135	Fields Of The World	2024	S,I	2/3	S2 MSI	70.8K	256	10m	—	Yes	Yes		GB2+TG
136	GAMUS	2024	S	7	RGB+nDSM	—	—	1m	10	Yes	?		Paper
137	Kuro Siwo	2024	S	3	S1 SAR+DEM	67.5K	224	10m	—	Yes	Yes		CB+GB2+Paper
138	MADOS	2024	S	15	S2 MS	2.8K	40~240	10~60m	5.35	Yes	Yes		PG
139	DL4GAM	2025	S	2	S2 MSI	2.3K/11.4K	256	10m	—	Yes	?		TG
140	CropSeg	2026	S	—	HLS MS	3,854	—	30m	—	Yes	Yes		GFM
141	FLAIR-HUB	2026	S	19/23	Multi	241.1K	10~512	0.2~10m	—	Yes	?		TG
142	CaFFe	—	S	2/4	SAR	19.1K	512	6~20m	—	Yes	Yes		GB2+TG
143	EnviroAtlas	—	S	10	Multi	—	—	1m	—	Yes	?		TG
144	L7 Irish	—	S	5	L7 MSI	206	7.5K~8.4K	15~30m	—	Yes	?		TG
145	Solar Plants Brazil	—	S	2	RGB+NIR	272	256	10m	—	Yes	?		TG
146	Substation	—	S	2	S2 MSI	27K	228	10m	—	Yes	Yes		GB2+TG
147	OSCD	2018	CD	2	S2	24	600	10m	0.48	Yes	Yes		GFM+TG
148	xView2	2018	CD	4	RGB	22.1K	1024	0.5m	51	Yes	Yes		PG+TG
149	LEVIR-CD	2020	CD	1	RGB	637	1024	0.5m	2.7	Yes	Yes		GFM+TG
150	DFC21-MSD	2021	CD	15	MS+MT	2.2K×9	4000	1~30m	325	Yes	?		Paper
151	Dynamic EarthNet Challenge	2021	CD	7	RGB	22.5K	1024	3m	—	Yes	?		Name
152	RapidAI4EO	2021	CD	44	MS+RGB	500K	—	3m,10m	—	Yes	Yes		FOMO
153	WH-MAVS	2021	CD	15	RGB	47.1K	200	1.2m	—	Yes	?		Paper
154	BRIGHT	2025	CD	4	RGB+SAR	3.2K	1024	0.1~1m	—	Yes	?		TG
155	FAIR-EO CD Benchmark	2026	CD	—	Multi	—	—	—	—	Yes	?		Name
156	CaBuAr	—	CD	2	S2 MSI	424	512	20m	—	Yes	?		TG
157	ChaBuD	—	CD	2	S2 MSI	356	512	10m	—	Yes	?		TG
158	NWPU-VHR10	2014	OD	10	RGB+IRRG	800	1000	0.08~2m	0.073	Yes	?		TG
159	COWC	2016	OD	1	RGB	388.4K	256	0.15m	64	Yes	?		TG
160	Airbus Ship Detection Challenge	2018	OD	—	RGB	192.6K	768	—	30	Yes	?		Name+Paper
161	DOTA	2018	OD	15	RGB	5.2K	800~4K	—	—	Yes	?		TG
162	Map Challenge	2018	OD	2	RGB	341.1K	300	—	—	Yes	?		Name
163	TBF	2018	OD	2	RGB	13	0~40	—	—	Yes	?		Paper
164	VisDrone2019-DET	2019	OD	10	RGB	10.2K	2000x1500	—	2	Yes	?		Paper
165	VisDrone2019-MOT	2019	OD	5	RGB Video	40K	3840x2160	—	14	Yes	?		Paper
166	VisDrone2019-SOT	2019	OD	3	RGB Video	139.3K	3840x2160	—	68	Yes	?		Paper
167	VisDrone2019-VID	2019	OD	5	RGB	40K	3840x2160	—	14	Yes	?		Paper
168	DIOR	2020	OD	20	RGB	23.5K	800	0.5m-30m	—	Yes	Yes		REO+TG

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Index	Dataset	Year	Task	# Cls.	Sensor / modality	# Samples	Size (px)	Resolution (m)	GB	Benchmark	FM Eval	Evidence
169	FAIR1M	2020	OD	37	OPT (Satellite)	15.3K	5000	–	120	Yes	?	TG
170	MOR-UAV	2020	OD	2	RGB	10.9K	608	–	–	Yes	?	Paper
171	RSOC	2020	OD	4	RGB	3.1K	2500	–	–	Yes	?	Paper
172	ArtifiVe-Potsdam	2021	OD	1	MS	4.8K	600	0.05m	15.6	Yes	?	Paper
173	Forest Damage	2021	OD	5	RGB	1.5K	1500	–	3.3	Yes	?	TG
174	NASADEbris	2021	OD	1	OPT (Satellite)	707	256	–	0.073	Yes	?	TG
175	NEON Tree Dataset	2021	OD	1	Multiple (Aerial)	2.2K×426	40	–	8.4	Yes	Yes	FOMO
176	SeaDronesSee	2022	OD	6	RGB	5.6K	3~840	–	60.3	Yes	?	Paper
177	SODA	2022	OD	9	RGB	2.5K	2.7K~4.8K	–	–	Yes	?	TG
178	UAVOD-10	2022	OD	10	OPT	844	–	–	–	Yes	?	Paper
179	EverWatch	2024	OD	8	RGB	5.3K	1500	0.01m	–	Yes	Yes	GB2+TG
180	VRSBench	2024	IC	–	OPT+Text	29.6K	–	–	–	Yes	Yes	REO
181	SEN12MS-CR	2020	SR	–	S1/S2	122.2K triplets	256	10m	–	Yes	Yes	GFM+Paper
182	WorldStrat	2022	SR	8	Multi	3.4K	1054	1.5m-60m	110	Yes	?	TG
183	GEO-Bench	2023	B	–	Multi	12 datasets	–	–	~65	Yes	Yes	GB1+Name
184	FoMo-Bench	2024	B	–	Multi	15 datasets	–	–	–	Yes	Yes	FOMO+Name
185	PANGAEA	2024	B	–	Multi	11 datasets	–	–	–	Yes	Yes	Name+PG
186	Copernicus-Bench	2025	B	–	Multi	15 datasets	–	–	–	Yes	Yes	CB+Name
187	PlantationBench	2025	B	–	Multi	–	–	–	–	Yes	Yes	Name+PB
188	EarthShift	2026	B	–	Multi	11 tasks	–	–	–	Yes	Yes	ES+Name
189	MMEarth-Bench	2026	B	–	Multi	5 tasks	–	–	–	Yes	Yes	MMB+Name

Subset: only entries with **Benchmark = Yes** are included. *FM Eval:* Yes = documented use for foundation-model / GeoFM evaluation; ? = not yet established. *Evidence:*

TG = TorchGeo; GB1 = GEO-Bench; GB2 = GEO-Bench-2; GFM = documented GeoFM/FM evaluation; PG = PANGAEA; FOMO = FoMo-Bench; CB = Copernicus-Bench;

REO = REOBench; LB = Landsat-Bench; PH = PhilEO Bench; PB = PlantationBench; ES = EarthShift; MMB = MMEarth-Bench; Paper = scientific paper; Name = explicit

benchmark/challenge/evaluation naming.